KEYWAVE

KW007: Ultra-Low Power 5.8GHz Microwave Radar Sensor

Motion Detection with Intelligent Doppler Engine

Programming Guide

Version 2.0c

Aug 2026

KEYWAVE

KW007: Ultra-Low Power 5.8GHz Microwave Radar Sensor

General Description:

The **KW007** is a **Doppler radar sensor** designed for ease of use; detection results can be **read from pin** directly, or the device can be configured and monitored via an **I²C interface**.

Once configured, the device operates autonomously, detecting object motion relative to the sensor and updating status flags to indicate **approaching** or **leaving** targets.

Key Operational Characteristics:

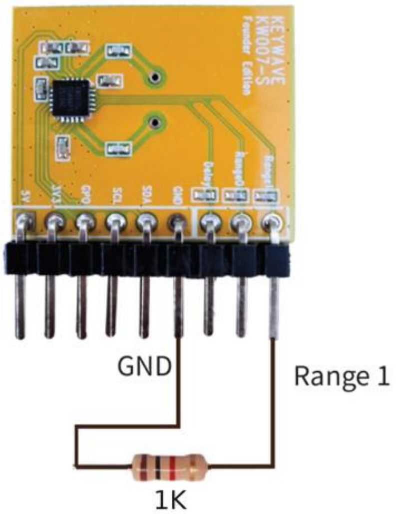
- The internal status flags (Approach/Leaving) are updated at an approximate rate of **8 Hz**.
- Hardware flags can be read directly via **I²C** for low-power **MCU** designs
- **VCO bank calibration** is factory-programmed and must be read during initialization
- Detection range is configurable through **amplifier gain** and **threshold** settings

Applications:

- **Smart Lighting:** Motion-triggered indoor/outdoor light control.
- **Touchless Control:** Hands-free faucets, soap dispensers, and automatic lids.
- **Presence Detection:** Proximity sensing for kiosks, digital signage, and smart displays.
- **Device Wake-up:** Low-power "wake-on-approach" for IoT devices and security panels.
- **Home Automation:** Context-aware sensing for smart home ecosystems.

Hardware Note

When using KW007 in **I²C mode,**
<u>the module's Range1 pin must be connected to GND through a resistor of approximately 1 kΩ</u>.
(or driven low by the MCU through the same resistor)



KEYWAVE

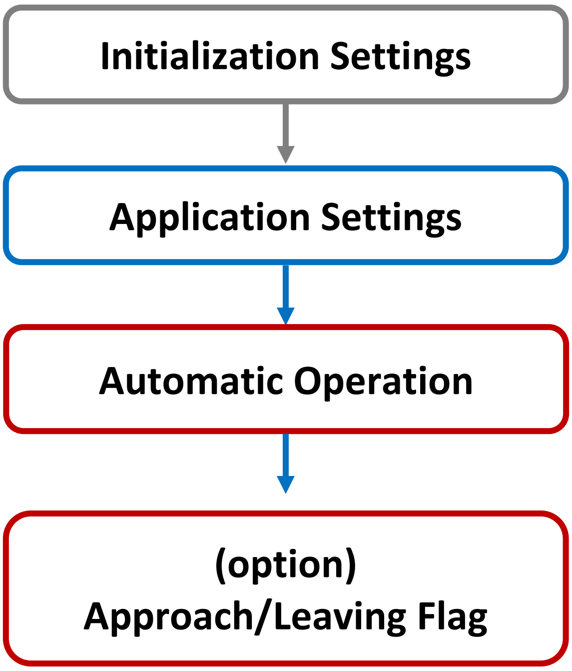
Power Supply Note

Use either the 3.3 V or the 5 V input, never both at the same time, and never apply 5 V to the 3.3 V input — doing so will damage the module. 3.3 V is the recommended supply. With either rail, decouple close to the module and provide transient and surge protection: supply ripple couples into the receiver, reducing effective range and increasing false triggers.

KW007 Doppler Radar Software Procedure

The **KW007** Doppler radar sensor can operate via a standard **I²C interface**. Upon power-up, the device must perform the following initialization sequence to ensure stable operation and optimal detection performance.

1. **Initialization Settings:** Write the fundamental chip configuration parameters to initialize the device.
2. **Application Settings:** Configure the sensing parameters based on specific application requirements. This includes setting the detection range and Doppler filter corners.
3. **Automatic Operation:** Once the configuration is applied, the radar operates automatically. The GPO hardware pin will show radar detection result.
4. **(Option) Approach/Leaving Flag:** The host MCU can poll the radar status at a fixed interval for further application processing.



Standard I²C Write/Read Interface Description for KW007

Supported Modes: Standard-mode (100 kHz) / Fast-mode (400 kHz)
Device Address: 7-bit addressing
Data Format: 8-bit Data, MSB First
Bus Pull-ups: SDA and SCL require pull-up resistors (typ. 4.7 kΩ)

KW007 Write/Read Address

Every transfer begins with the KW007's fixed 7-bit target address, followed by a read/write (R/W) bit. Together these form the first byte on the bus.

- **Slave address ID:** The 7-bit slave address ID is **0011001b**(0x19).
  With the R/W bit appended, the transmitted byte is
  0x32 for write and
  0x33 for read

KW007 Write mode Write format is shown as follow:

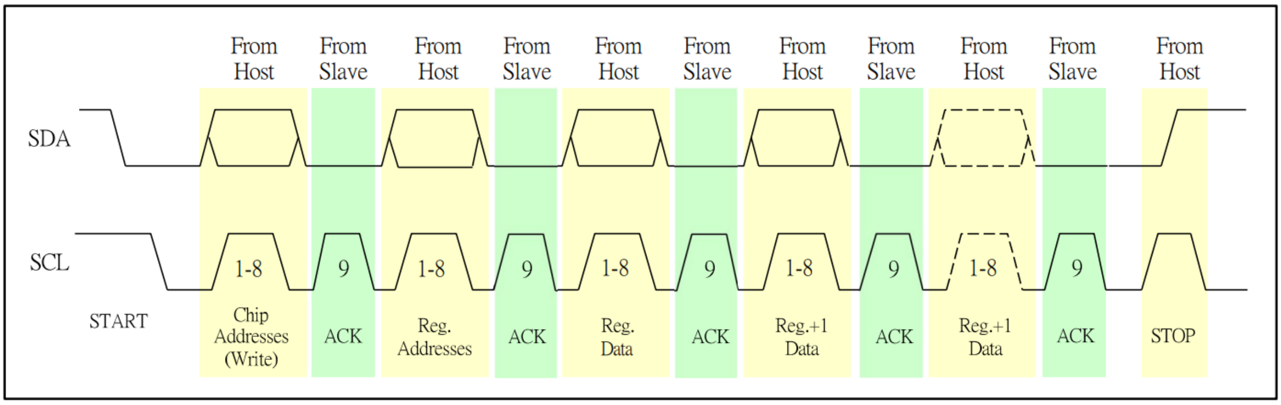


Figure 1. KW007 I²C Write Mode Timing Diagram

KW007 Read mode Read format is shown as follow:

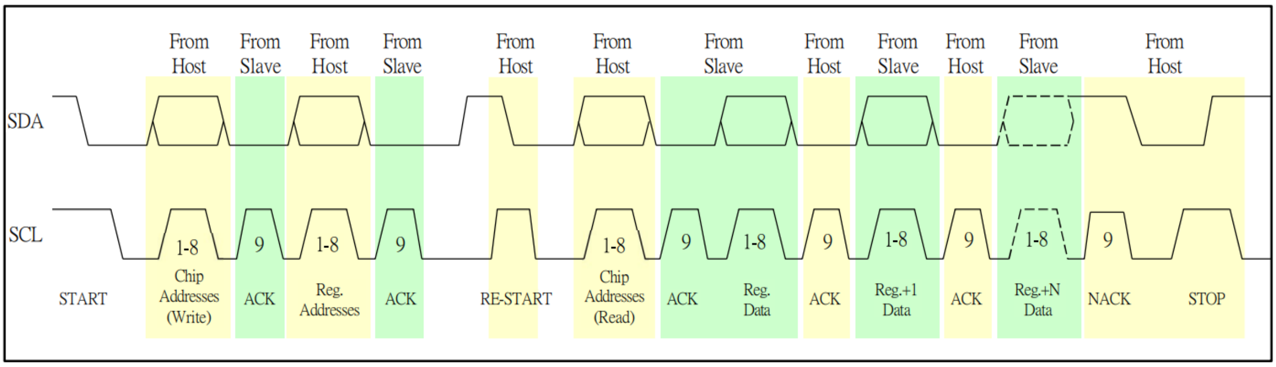


Figure 2. KW007 I²C Read Mode Timing Diagram

Register Descriptions

<table>
<tr><th>Register</th><th>Function</th><th>Access</th><th>Value</th><th>Bits</th><th colspan="2">Description</th></tr>
<tr><td>Reg 0</td><td>R0</td><td>R</td><td>-</td><td>[7:0]</td><td colspan="2">Reserved</td></tr>
<tr><td>Reg 1</td><td>VCO Bank</td><td>R</td><td>-</td><td>[5:0]</td><td colspan="2">VCO Bank (Read Only, factory burned)</td></tr>
<tr><td>Reg 2
|
Reg 4</td><td>R2
|
R4</td><td>R</td><td>-</td><td>[7:0]</td><td colspan="2">Reserved</td></tr>
<tr><td>Reg 5</td><td>RSSI</td><td>R</td><td>-</td><td>[7:0]</td><td colspan="2">RSSI Output (Read Only, 0-255)</td></tr>
<tr><td rowspan="3">Reg 6</td><td rowspan="3">Detection Flag: Day/Night & Approach & Leaving</td><td rowspan="3">R</td><td rowspan="3">-</td><td>[3]</td><td colspan="2">Day/Night Flag
0: Night
1: Day</td></tr>
<tr><td>[2]</td><td colspan="2">Flag Forward/Approach
0: No object approaching
1: Object approaching</td></tr>
<tr><td>[1]</td><td colspan="2">Flag Backward/Leaving
0: No object leaving
1: Object leaving</td></tr>
<tr><td>Reg 7</td><td>R7</td><td>R</td><td>-</td><td></td><td colspan="2">Reserved</td></tr>
<tr><td>Reg 8</td><td>R8</td><td>W/R</td><td>0xFC</td><td>[7:0]</td><td colspan="2">Reserved</td></tr>
<tr><td>Reg 9</td><td>Manual Mode</td><td>W/R</td><td>0x30</td><td>[5]</td><td colspan="2">[5]: Efuse mode
0: FUSE
1: I²C</td></tr>
<tr><td>Reg 10</td><td>R10</td><td>W/R</td><td>0x88</td><td>[7:0]</td><td colspan="2">Reserved</td></tr>
<tr><td>Reg 11</td><td>R11</td><td>W/R</td><td>0x03</td><td>[7:0]</td><td colspan="2">Reserved</td></tr>
<tr><td rowspan="2">Reg 12</td><td rowspan="2">Sampling Rate</td><td>W/R</td><td rowspan="2">0x54</td><td>[7]</td><td colspan="2">Sampling Rate: 0=1kHz, 1=500Hz</td></tr>
<tr><td>W/R</td><td>[5:0]</td><td colspan="2">VCO Bank (Set during initialization)</td></tr>
<tr><td>Reg 13
|
Reg 18</td><td>R13
|
R18</td><td>W/R</td><td>-</td><td>[7:0]</td><td colspan="2">Reserved</td></tr>
<tr><td>Reg 19</td><td>Amp1_Gain</td><td>W/R</td><td>0xEA</td><td>[7:5]</td><td>Amplifier 1 Gain:
000: 8 dB
001: 12 dB
010: 16 dB
011: 19 dB
100: 21 dB
101: 24 dB
110: 28 dB
111: 32 dB</td><td rowspan="2">Total Gain
= Amp1_Gain + Amp2_Gain + Amp3_Gain

(Ex. 24+24+0 = 48db)</td></tr>
<tr><td>Reg 20</td><td>Amp3_Gain</td><td>W/R</td><td>0xF7</td><td>[4:3]</td><td>Amplifier 3 Gain:
00: 0 dB
01: 4 dB
10: 8 dB
11: 12 dB</td></tr>
</table>

Register	Function	Access	Value	Bits	Description
	Amp2_Gain	W/R		[2:0]	Amplifier 2 Gain: 000: 8 dB 001: 12 dB 010: 16 dB 011: 19 dB 100: 21 dB 101: 24 dB 110: 28 dB 111: 32 dB
Reg 21	Slow Speed Detection	W/R	0x5F	[5]	High-pass Corner: (Default = 0) 0: Low (Allows slow-moving signals.) 1: High (Cuts off slow-moving signals.)
Reg 22 | Reg 23	R22 | R23	W/R	-	[7:0]	Reserved
Reg 24	Enhanced Mode	W/R	0xA0	[6]	Enhanced Range Mode: 0: Off (Normal mode) 1: On (Enhanced mode)
Reg 25 | Reg 26	R25 | R26	W/R	-	[7:0]	Reserved
Reg 27	Valid Decision Magnitude Threshold	W/R	0x30	[7:0]	Decision Magnitude Threshold 00000000: 0 00000001: 1 00000010: 2 … 11111111: 255
Reg 28	R28	W/R	0x55	[7:0]	Reserved
Reg 29	GPO Delay	W/R	0x75	[2:1]	GPO Delay: (Default = 10) SE/L S (Model) 00: 16s 4s 01: 32s 2s 10: 1s 1s 11: 64s 64s
	High Speed Detection	W/R		[0]	Low-pass Corner: (Default = 1) 0: Low (Filters out ultra-fast signals.) 1: High (allow more fast objects detect.)
Reg 30 | Reg 31	R30 | R31	W/R	-	[7:0]	Reserved

KW007 Recommended Register Configuration:

Address	Value(Hex)	Address	Value(Hex)	Address	Value(Hex)	Address	Value(Hex)
Reg 0	0x00	Reg 8	0xFC	Reg 16	0x98	Reg 24	0xA0
Reg 1	0x00	Reg 9	0x30	Reg 17	0xA0	Reg 25	0x20
Reg 2	0x00	Reg 10	0x88	Reg 18	0x00	Reg 26	0x9A
Reg 3	0x00	Reg 11	0x03	Reg 19	0xEA	Reg 27	0x30
Reg 4	0x00	Reg 12	0x54	Reg 20	0xF7	Reg 28	0x55
Reg 5	0x00	Reg 13	0x4B	Reg 21	0x5F	Reg 29	0x75
Reg 6	0x00	Reg 14	0x68	Reg 22	0x02	Reg 30	0x42
Reg 7	0x00	Reg 15	0x55	Reg 23	0x30		

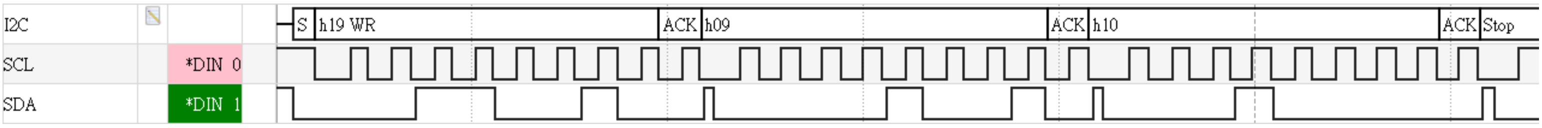
Note: Reg9 = 0x30 (manual mode), Reg12[5:0] will be set to VCO bank value.

Initialization Flow

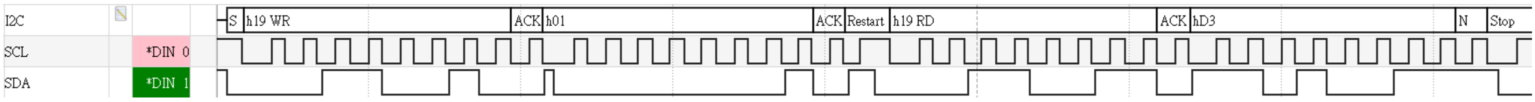
Follow these steps in order:

1. **Write REG9 = 0x10** (Efuse mode) to enable reading the factory-burned VCO bank value
2. **Read REG1[5:0]** to obtain the VCO bank value (factory calibrated for each chip)
3. **Set REG12[5:0] = VCO bank value** in your initialization register array
4. **Write all 31 registers (R0-R30)** with REG9 = 0x30 (I^2C manual mode)

Write Reg9 = 0x10

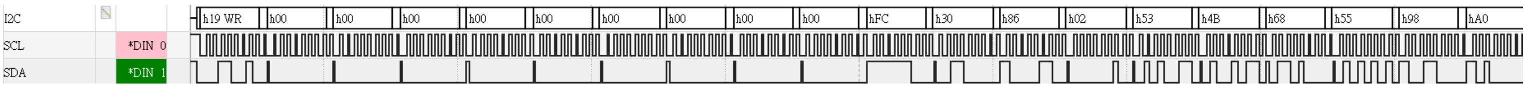


Read Reg1[5:0]



Set Reg12[5:0] = VCO bank

Write all 31 Registers



KEYWAVE

Functional Description

Detection Range

The detection range depends on two key parameters:

- Amplifier Gain: Increasing the gain extends the range.
- Magnitude Threshold: Lowering the threshold extends the range.

While higher gain or lower thresholds increase range, they also heighten susceptibility to false triggers.

Amplifier Gain Configuration

The total system gain is determined by summing the gain values of the three amplifier stages:

$$Total\ Gain = Amp_1\ Gain + Amp_2\ Gain + Amp_3\ Gain$$

Gain Selection Table

Bit Value	Amp_1 Gain (dB)	Amp_2 Gain (dB)	Amp_3 Gain (dB)
000	8	8	0
001	12	12	4
010	16	16	8
011	19	19	12
100	21	21	-
101	24	24	-
110	28	28	-
111	32	32	-

For optimal Signal-to-Noise Ratio (SNR), it is recommended to apply higher gain settings in the earlier stages ($Amp_1 > Amp_2 > Amp_3$).

Amp3 Gain is a two-bit field: use 00, 01, 10 and 11 for 0, 4, 8 and 12 dB.

RSSI Readout

Reg5 reports the strength of the reflected signal after the three-stage amplifier, as a single read-only byte from 0 to 255. Suggest to add a software digital filter after readout to smooth the result.

Once testing and setup are complete, it does not need to be read during normal operation.

Magnitude Threshold Configuration

A target is detected only when the signal magnitude exceeds this threshold. For optimal results, follow the recommended setup sequence below:

1. Measure Baseline RSSI

   The "Received Signal Strength Indicator" (RSSI) is the strength of the reflected signal: it rests at the background level when no target is present and rises when one moves within range. The RSSI chart in the lower-right panel of the GUI shows it live.

2. Set "Mag Threshold" Above Noise Floor
3. Refer to the live RSSI to tune "Mag Threshold": set it above the noise floor, but below the level reached when a target moves at the farthest distance that must still be detected. Changing the amplifier gain moves both readings, so re-check the threshold afterwards.

Note:

- Signal Saturation:

  Improper gain causes signal saturation (clipping at 0 or 255), leading to detection failure.

- False Triggers:

  If "Mag Threshold" is too close to baseline RSSI, environmental noise will trigger false detections.

- "Enhanced Range Mode" uses optimized parameters to boost total detection distance.

Doppler Filter Corner

This section configures the corner frequencies for the Doppler filters:

Filter Type	Function	Trade-off / Impact
Low-Pass Filter (LPF)	Attenuate high-speed clutter and high-frequency noise	Setting too low may attenuate fast-walking targets
High-Pass Filter (HPF)	Reject slow-motion background clutter	Setting too high may miss subtle, slow human movements

Sampling Rate

Key considerations for sampling rate adjustment:

- Filter Scaling:

  The Doppler filter corner frequencies scale proportionally with the sampling rate.

- Noise Reduction:

  A higher sampling rate mitigates noise aliasing, effectively lowering the in-band background noise floor.

GPO Delay

Specifies the hold time for GPO.

- Idle(GPO low): Default state when no motion is detected.
- Trigger(GPO active high): GPO is triggered instantly upon motion detection.
- Release: GPO is held active after motion stops, returning to Idle after delay expires.

Approach Flag

The Reg6[2] defines whether an object is moving toward the radar.

Read only.

- 0: No object approaching.
- 1: Object approaching.

Leaving Flag

The Reg6[1] defines whether an object is moving away from the radar.

Read only.

- 0: No object leaving.
- 1: Object leaving.

Day/Night Flag

The Reg6[3] defines ambient light condition.

Read only.

- 0: Night.
- 1: Day.

Recommended Settings Table

Detection Range (Typical Estimate)	Total Gain Reg19[7:5] + Reg20[2:0] + Reg20[4:3] = AMP1 + AMP2 + AMP3	Magnitude Threshold Reg27[7:0]	Enhanced Mode Reg24[6]
Level 1 (~15 cm)	**19+8+0 = 27 dB**	**45**	OFF
Level 2 (~50cm)	**24+8+0 = 32dB**	**64**	OFF
Level 3 (~1 m)	**32+12+0 = 44 dB**	**90**	OFF
Level 4 (~3 m)	**32+21+0 = 53 dB**	**90**	OFF
Level 5 (~5 m)	**32+32+0 = 64 dB**	**90**	OFF
Level 6 (~7 m)	**32+32+4 = 68 dB**	**64**	OFF
Level 7 (~10 m)	**32+32+12 = 76 dB**	**32**	OFF
Level 8 (~13 m)	**32+32+12= 76 dB**	**32**	**ON**

LICENSE & ATTRIBUTION

Revision History	Date	Description
V2.0c	2026-08	Rewrite I²C addressing description; correct Day/Night flag, Amp3_Gain encoding and Total Gain formula; add RSSI Readout, Mag Threshold adjustment, power supply and hardware connection notes.
V1.9	2026-04	Modify Register Table
V1.8	2026-04	Modify Settings Table
V1.7	2026-03	Modify Reg9 Description Update Recommended Table (Level 1~8)
V1.6	2026-03	Reg29 0x65 -> 0x75

End of Document